

Project 10 – Cloud workload execution models

A cloud computing infrastructure provides a total computing capacity of C GHz and receives requests for workload execution from N users. Each user generates a new workload every T seconds, and each workload requires a computing capacity of R GHz for D seconds. T , R and D are IID random variables whose distribution is specified below.

Each user submits its workloads to the cloud infrastructure according to one of the following execution models:

- *Container-based model*: each workload is executed within a long-term container, which runs at R GHz for D seconds. The container takes I_{cont} seconds to be instantiated and I_{cont} seconds to be terminated.
- *Function as a Service (Faas) model*: each workload is divided into K functions of equal duration, and functions are executed sequentially, i.e., function k is started only after function $k-1$ has completed. Each function is executed within a short-term container that runs at R GHz for the time needed to execute that function only. The container takes I_{Faas} seconds to be instantiated and I_{Faas} seconds to be destroyed. Moreover, $I_{Faas} < I_{cont}$.

For each incoming workload or function, the cloud infrastructure checks if there is enough capacity to execute it immediately. If capacity is insufficient, the workload or function is placed into a First Come First Served (FCFS) queue and starts execution as soon as enough capacity becomes available.

Evaluate at least the total execution time of workloads for various values of N and K for the two models in the following scenario(s):

- exponential distribution for T and D , uniform distribution of R .

In all cases, it is up to the team to calibrate the scenarios so that meaningful results are obtained.

Project deliverables:

- a) Documentation (according to the standards set during the lectures)
- b) Simulator code
- c) Presentation (up to 10 slides maximum)